

MYSQL VIEWS

Practical Examples and Student Exercises using MySQL Workbench

2520030396

Mokshitha putti

1. What is a View?

A **View** is a **virtual table** created from one or more existing tables.

It **does not store data** itself.

It stores only the **SQL query**.

Whenever you query the view, MySQL retrieves the latest data from the underlying tables.

2. Setup: Employees and Departments Tables

Run this setup first. If these tables already exist with data, you may skip the setup.

```
CREATE DATABASE IF NOT EXISTS employeeedb1;
```

```
USE employeeedb1;
```

```
CREATE TABLE IF NOT EXISTS departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE IF NOT EXISTS employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary DECIMAL(10,2)  
);
```

```
INSERT IGNORE INTO departments VALUES  
(1,'CSE'),  
(2,'ECE'),  
(3,'EEE');
```

```
INSERT IGNORE INTO employees VALUES  
(101,'Rahul',1,50000),  
(102,'Priya',2,65000),  
(103,'Anil',1,55000),  
(104,'Sneha',3,70000),  
(105,'Kiran',2,48000);
```

```

Query 1 x
1 • SHOW TABLES;
2 • SELECT * FROM departments;
3 • SELECT * FROM employees;
4

```

Result Grid				
	emp_id	emp_name	dept_id	salary
▶	101	Rahul	2	53000.00
	102	Priya	2	67000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00
+	ROLL	ROLL	ROLL	ROLL

Result 1 departments 2 employees 3 x			
Output			
Action Output			
#	Time	Action	Message
2	16:58:11	USE employeedb 1	0 row(s) affected
3	16:58:11	CREATE TABLE IF NOT EXISTS departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50))	0 row(s) affected, 1 warning(s): 1050 Table 'dep'
4	16:58:11	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), de...	0 row(s) affected, 1 warning(s): 1050 Table 'emg'
5	16:58:11	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'EEE')	0 row(s) affected, 3 warning(s): 1062 Duplicate
6	16:58:11	INSERT IGNORE INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), ...	0 row(s) affected, 5 warning(s): 1062 Duplicate
7	16:58:31	SHOW TABLES	2 row(s) returned
8	16:58:31	SELECT * FROM departments LIMIT 0, 1000	3 row(s) returned
9	16:58:31	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned

3. Verify the Tables

```

SHOW TABLES;
SELECT * FROM departments;
SELECT * FROM employees;

```

View 1 – Display All Employees

Question: Create a View named EmployeeView to display all columns from employees.

```

DROP VIEW IF EXISTS EmployeeView;
CREATE VIEW EmployeeView AS
SELECT *
FROM employees;
SELECT * FROM EmployeeView;

```

Query 1

```

1 DROP VIEW IF EXISTS EmployeeView;
2 CREATE VIEW EmployeeView AS
3 SELECT * FROM employees;
4
5 SELECT * FROM EmployeeView;
6

```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	2	53000.00
102	Priya	2	67000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

EmployeeView 4

Output

Action Output

#	Time	Action	Message
5	16:58:11	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'EEE')	0 row(s) af
6	16:58:11	INSERT IGNORE INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), ...	0 row(s) af
7	16:58:31	SHOW TABLES	2 row(s) re
8	16:58:31	SELECT * FROM departments LIMIT 0, 1000	3 row(s) re
9	16:58:31	SELECT * FROM employees LIMIT 0, 1000	5 row(s) re
10	16:59:13	DROP VIEW IF EXISTS EmployeeView	0 row(s) af
11	16:59:13	CREATE VIEW EmployeeView AS SELECT * FROM employees	0 row(s) af
12	16:59:13	SELECT * FROM EmployeeView LIMIT 0, 1000	5 row(s) re

View 2 – Selected Columns

Question: Create a View showing employee ID, employee name and salary only.

DROP VIEW IF EXISTS EmployeeBasicView;

CREATE VIEW EmployeeBasicView AS

SELECT emp_id, emp_name, salary

FROM employees;

SELECT * FROM EmployeeBasicView;

Query 1

```

2 CREATE VIEW EmployeeBasicView AS
3 SELECT emp_id, emp_name, salary
4 FROM employees;
5
6 SELECT * FROM EmployeeBasicView;
7

```

Result Grid

emp_id	emp_name	salary
101	Rahul	53000.00
102	Priya	67000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

EmployeeBasicView 5

Output

Action Output

#	Time	Action	Message
8	16:58:31	SELECT * FROM departments LIMIT 0, 1000	3 row(s) returned
9	16:58:31	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned
10	16:59:13	DROP VIEW IF EXISTS EmployeeView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeeb1.e
11	16:59:13	CREATE VIEW EmployeeView AS SELECT * FROM employees	0 row(s) affected

View 3 – High Salary Employees

Question: Create a View that displays employees earning more than 55000.

DROP VIEW IF EXISTS HighSalaryEmployees;

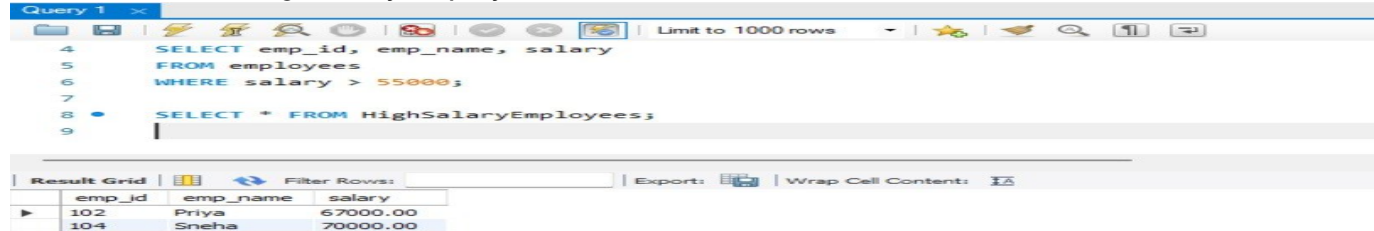
CREATE VIEW HighSalaryEmployees AS

SELECT emp_id, emp_name, salary

FROM employees

WHERE salary > 55000;

SELECT * FROM HighSalaryEmployees;

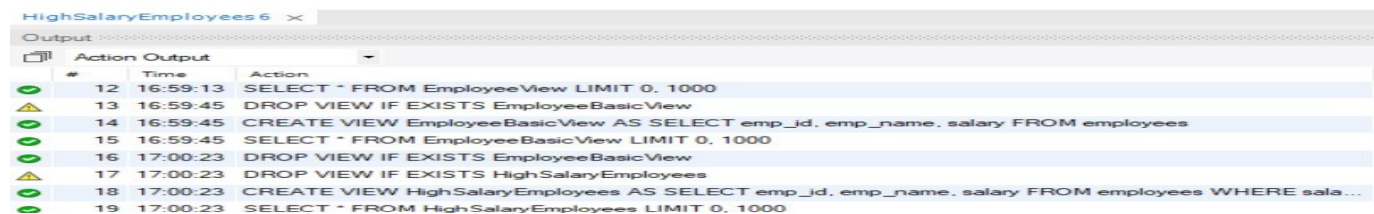


Query 1

```
4 SELECT emp_id, emp_name, salary
5 FROM employees
6 WHERE salary > 55000;
7
8 SELECT * FROM HighSalaryEmployees;
9
```

Result Grid

emp_id	emp_name	salary
102	Priya	67000.00
104	Sneha	70000.00



HighSalaryEmployees 6

Output

#	Time	Action
12	16:59:13	SELECT * FROM EmployeeView LIMIT 0, 1000
13	16:59:45	DROP VIEW IF EXISTS EmployeeBasicView
14	16:59:45	CREATE VIEW EmployeeBasicView AS SELECT emp_id, emp_name, salary FROM employees
15	16:59:45	SELECT * FROM EmployeeBasicView LIMIT 0, 1000
16	17:00:23	DROP VIEW IF EXISTS EmployeeBasicView
17	17:00:23	DROP VIEW IF EXISTS HighSalaryEmployees
18	17:00:23	CREATE VIEW HighSalaryEmployees AS SELECT emp_id, emp_name, salary FROM employees WHERE sala...
19	17:00:23	SELECT * FROM HighSalaryEmployees LIMIT 0, 1000

View 4 – Employee and Department using JOIN

Question: Create a View that displays employee name, department name and salary.

DROP VIEW IF EXISTS EmployeeDepartmentView;

CREATE VIEW EmployeeDepartmentView AS

SELECT

e.emp_id,

e.emp_name,

d.dept_name,

e.salary

FROM employees e

JOIN departments d

ON e.dept_id = d.dept_id;

SELECT * FROM EmployeeDepartmentView;

Query 1 x

Limit to 1000 rows

```
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id;
12
13 SELECT * FROM EmployeeDepartmentView;
14
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	ECE	53000.00
	102	Priya	ECE	67000.00
	103	Anil	CSE	55000.00
	104	Sneha	EEE	70000.00
	105	Kiran	ECE	48000.00

Result Grid
Form Editor
Field Types

EmployeeDepartmentView 8 x Read Only Co

Output

View 5 – Employees from a Particular Department

Question: Create a View that displays only CSE employees.

DROP VIEW IF EXISTS CSEEmployees;

CREATE VIEW CSEEmployees AS

SELECT

e.emp_id,
e.emp_name,
d.dept_name,
e.salary

FROM employees e

JOIN departments d

ON e.dept_id = d.dept_id

WHERE d.dept_name = 'CSE';

SELECT * FROM CSEEmployees;

Query 1

```
10 JOIN departments d
11 ON e.dept_id = d.dept_id
12 WHERE d.dept_name = 'CSE';
13
14 • SELECT * FROM CSEEmployees;
15
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents:

	emp_id	emp_name	dept_name	salary
▶	103	Anil	CSE	55000.00

CSEEmployees 9 x Read Only

Output

View 6 – Department-wise Average Salary

Question: Create a View using GROUP BY and AVG() to display department-wise average salary.

```
DROP VIEW IF EXISTS DepartmentSalaryView;
CREATE VIEW DepartmentSalaryView AS
SELECT
    d.dept_name,
    COUNT(e.emp_id) AS EmployeeCount,
    AVG(e.salary) AS AverageSalary
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name;
SELECT * FROM DepartmentSalaryView;
```

Query 1

```

9      JOIN departments d
10     ON e.dept_id = d.dept_id
11     GROUP BY d.dept_id, d.dept_name;
12
13 •   SELECT * FROM DepartmentSalaryView;
14

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents:

	dept_name	EmployeeCount	AverageSalary
▶	ECE	3	56000.000000
	CSE	1	55000.000000
	EEE	1	70000.000000

DepartmentSalaryView 10 | Read Only

View 7 – Departments with Average Salary Above 55000

Question: Create a View that displays departments whose average salary is greater than 55000.

DROP VIEW IF EXISTS HighAverageDepartments;

CREATE VIEW HighAverageDepartments AS

SELECT

 d.dept_name,

 AVG(e.salary) AS AverageSalary

FROM employees e

JOIN departments d

ON e.dept_id = d.dept_id

GROUP BY d.dept_id, d.dept_name

HAVING AVG(e.salary) > 55000;

SELECT * FROM HighAverageDepartments;

Query 1

```

9      ON e.dept_id = d.dept_id
10     GROUP BY d.dept_id, d.dept_name
11     HAVING AVG(e.salary) > 55000;
12
13     SELECT * FROM HighAverageDepartments;
14

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents:

	dept_name	AverageSalary
▶	ECE	56000.000000
	EEE	70000.000000

HighAverageDepartments 11 | Read Only

View 8 – Salary Range View

Question: Create a View that displays employees with salary between 50000 and 70000.

Query 1

```
9      ON e.dept_id = d.dept_id
10     GROUP BY d.dept_id, d.dept_name
11     HAVING AVG(e.salary) > 55000;
12
13     SELECT * FROM HighAverageDepartments;
14
```

Result Grid

	dept_name	AverageSalary
▶	ECE	56000.000000
	EEE	70000.000000

HighAverageDepartments 11

Read Only

View 9 – View with Calculated Column

Question: Create a View that displays employee name, monthly salary and annual salary.

DROP VIEW IF EXISTS AnnualSalaryView;

CREATE VIEW AnnualSalaryView AS

SELECT

emp_id,

emp_name,

salary AS MonthlySalary,

salary * 12 AS AnnualSalary

FROM employees;

SELECT * FROM AnnualSalaryView;

Query 1

```
7      salary AS MonthlySalary,
8      salary * 12 AS AnnualSalary
9     FROM employees;
10
11 •   SELECT * FROM AnnualSalaryView;
12
```

Result Grid

	emp_id	emp_name	MonthlySalary	AnnualSalary
▶	101	Rahul	53000.00	636000.00
	102	Priya	67000.00	804000.00
	103	Anil	55000.00	660000.00
	104	Sneha	70000.00	840000.00
	105	Kiran	48000.00	576000.00

AnnualSalaryView 12

Read Only

View 10 – Update Data Through a Simple View

Question: Create a simple single-table View and update an employee's salary through the View.

```
DROP VIEW IF EXISTS EmployeeSalaryView;  
CREATE VIEW EmployeeSalaryView AS  
SELECT emp_id, emp_name, salary  
FROM employees;  
UPDATE EmployeeSalaryView  
SET salary = 60000  
WHERE emp_id = 101;  
SELECT * FROM EmployeeSalaryView;  
SELECT * FROM employees WHERE emp_id = 101;
```

The screenshot shows a database query tool interface. The top pane displays the SQL script for creating and updating a view. The bottom pane shows the result grid for the final query.

Query 1

```
8      SET salary = 60000  
9      WHERE emp_id = 101;  
10  
11 •   SELECT * FROM EmployeeSalaryView;  
12 •   SELECT * FROM employees WHERE emp_id = 101;  
13
```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	2	60000.00
•	NULL	NULL	NULL	NULL

EmployeeSalaryView 13 employees 14 x Apply

View 11 – CREATE OR REPLACE VIEW

Question: Modify an existing View so that it displays employee name and salary only.

```
CREATE OR REPLACE VIEW EmployeeBasicView AS  
SELECT emp_name, salary  
FROM employees;  
SELECT * FROM EmployeeBasicView;
```

Query 1 x

```

1 • CREATE OR REPLACE VIEW EmployeeBasicView AS
2   SELECT emp_name, salary
3   FROM employees;
4
5 • SELECT * FROM EmployeeBasicView;
6

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents:

	emp_name	salary
▶	Rahul	60000.00
	Priya	67000.00
	Anil	55000.00
	Sneha	70000.00
	Kiran	48000.00

EmployeeBasicView 15 x Read Only

Output

View 12 – View Definition

Question: Display the SQL statement used to create a View.

SHOW CREATE VIEW EmployeeDepartmentView;

Query 1 x

```

1 • SHOW CREATE VIEW EmployeeDepartmentView;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	View	Create View	charac
▶	employeedepartmentview	CREATE ALGORITHM=UNDEFINED DEFINER='r... utf8mb	

Result 16 x Read Only

Output

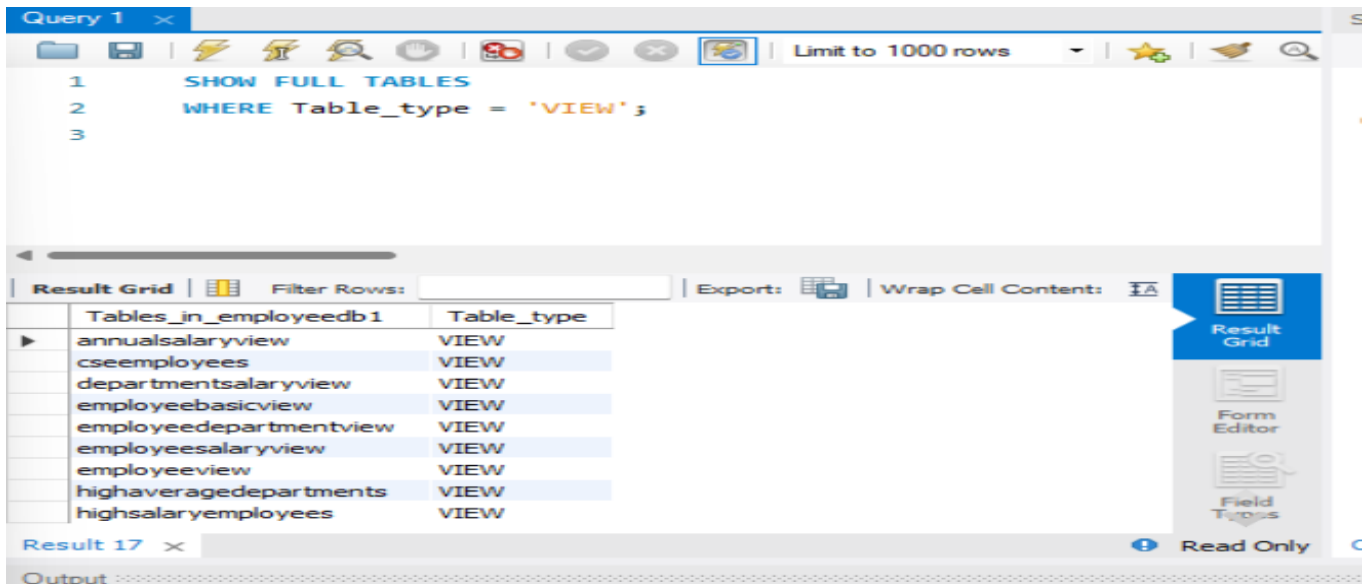
Action Output

View 13 – List All Views

Question: Display all Views available in the current database.

SHOW FULL TABLES

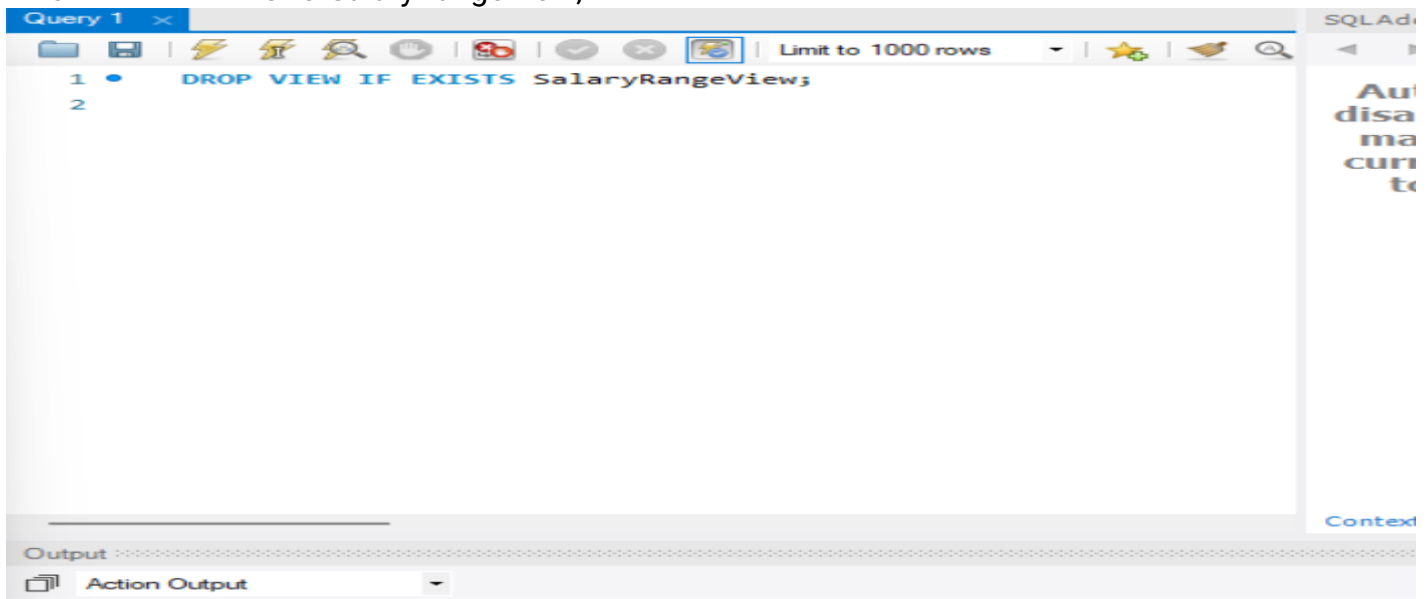
WHERE Table_type = 'VIEW';



View 14 – Drop a View

Question: Remove an existing View from the database.

DROP VIEW IF EXISTS SalaryRangeView;



4. Important Concept: Simple View vs Complex View

A simple View is generally based on one table and does not use grouping or aggregate functions. Some simple Views can be updated. A View containing JOINS, GROUP BY, aggregate functions such as AVG(), DISTINCT, or other non-updatable constructs may not be directly updatable.

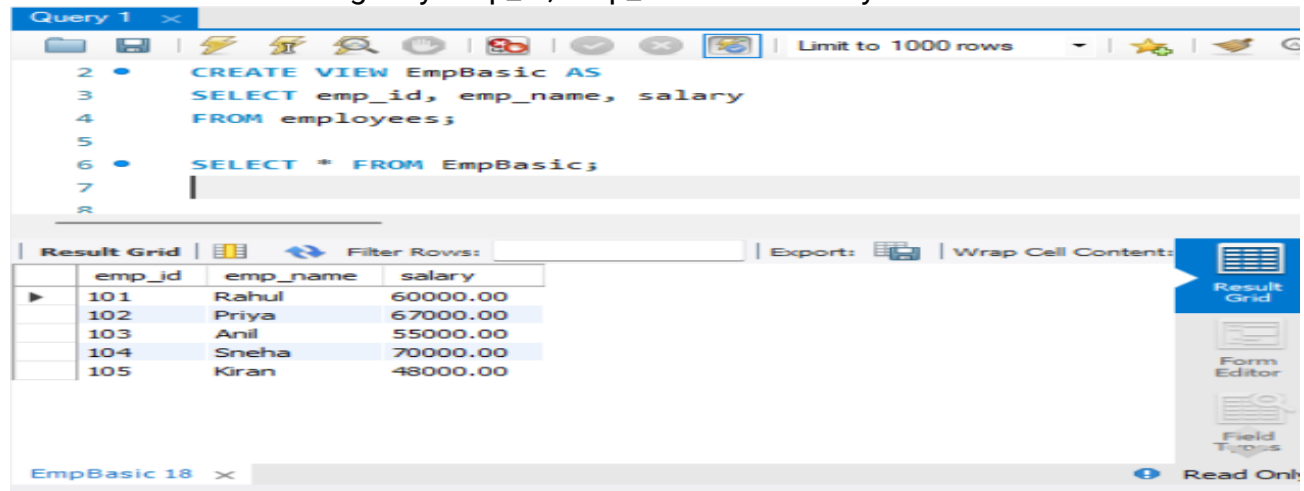
5. View vs Table

Feature	Table	View
Stores data	Yes	Normally no; stores query

Created with CREATE TABLE	Yes	definition
Created with CREATE VIEW	No	Yes
Can use JOINS	Not applicable	Yes
Reflects current underlying data	Own stored data	Yes
Can always be updated	Yes	No; depends on the View

6. Student Practice Tasks

1. Create a View showing only emp_id, emp_name and salary.



```

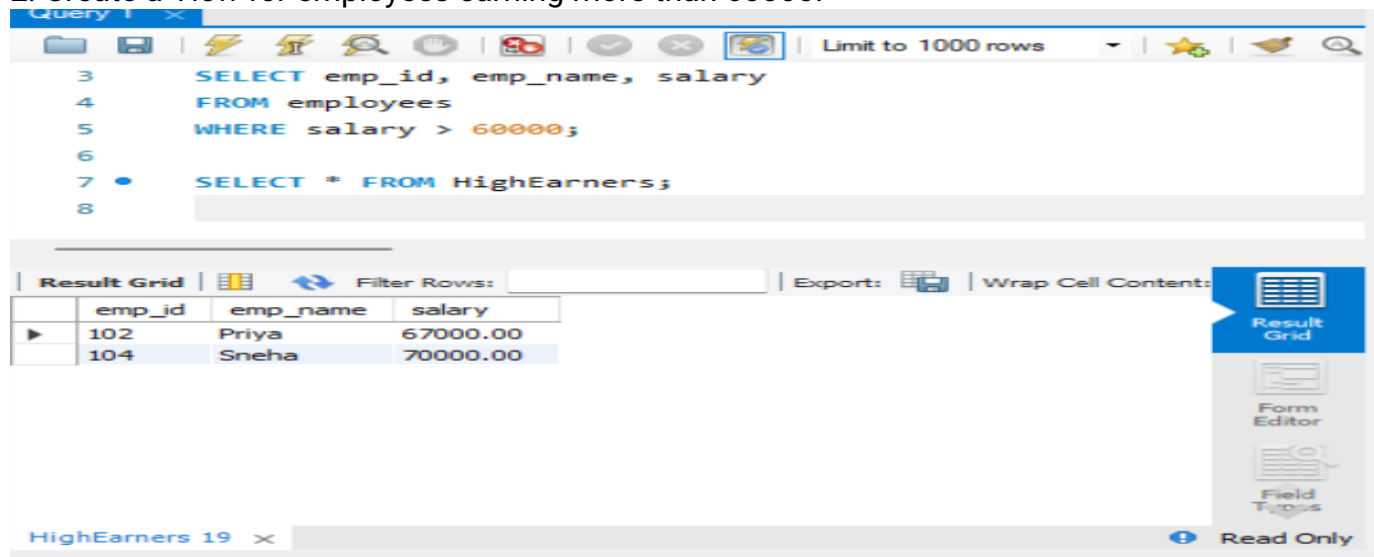
2 • CREATE VIEW EmpBasic AS
3   SELECT emp_id, emp_name, salary
4   FROM employees;
5
6 • SELECT * FROM EmpBasic;
7
8

```

emp_id	emp_name	salary
101	Rahul	60000.00
102	Priya	67000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

EmpBasic 18 x Read Only

2. Create a View for employees earning more than 60000.



```

3   SELECT emp_id, emp_name, salary
4   FROM employees
5   WHERE salary > 60000;
6
7 • SELECT * FROM HighEarnings;
8

```

emp_id	emp_name	salary
102	Priya	67000.00
104	Sneha	70000.00

HighEarnings 19 x Read Only

3. Create a View showing employee name, department name and salary using JOIN.

Query 1 x

```

3  SELECT e.emp_name, d.dept_name, e.salary
4  FROM employees e
5  JOIN departments d ON e.dept_id = d.dept_id;
6
7  SELECT * FROM EmpDept;
8

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name	salary
Rahul	ECE	60000.00
Priya	ECE	67000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	48000.00

EmpDept 20 x Read Only

4. Create a View showing only ECE employees.

Query 1 x

```

4  FROM employees e
5  JOIN departments d ON e.dept_id = d.dept_id
6  WHERE d.dept_name = 'ECE';
7
8  SELECT * FROM ECEEmployees;
9

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_name	salary
101	Rahul	ECE	60000.00
102	Priya	ECE	67000.00
105	Kiran	ECE	48000.00

ECEEmployees 21 x Read Only

5. Create a View showing department-wise employee count.

Query 1 x

```

4  FROM employees e
5  JOIN departments d ON e.dept_id = d.dept_id
6  GROUP BY d.dept_name;
7
8  SELECT * FROM DeptEmpCount;
9

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	EmployeeCount
ECE	3
CSE	1
EEE	1

DeptEmpCount 22 x Read Only

Output

6. Create a View showing department-wise maximum salary.

Query 1 x

```

1 • DROP VIEW IF EXISTS DeptMaxSalary;
2 CREATE VIEW DeptMaxSalary AS
3 SELECT d.dept_name, MAX(e.salary) AS MaxSalary
4 FROM employees e
5 JOIN departments d ON e.dept_id = d.dept_id
6 GROUP BY d.dept_name;
7

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents:

	dept_name	MaxSalary
▶	ECE	67000.00
	CSE	55000.00
	EEE	70000.00

DeptMaxSalary 23 x Read Only

Output

7. Create a View showing departments whose average salary is greater than 50000.

Query 1 x

```

5 JOIN departments d ON e.dept_id = d.dept_id
6 GROUP BY d.dept_name
7 HAVING AVG(e.salary) > 50000;
8
9 • SELECT * FROM DeptHighAvg;
10

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents:

	dept_name	AvgSalary
▶	ECE	58333.333333
	CSE	55000.000000
	EEE	70000.000000

DeptHighAvg 24 x Read Only

8. Create a View with employee name, salary and annual salary.

Query 1

```

2 CREATE VIEW EmpAnnualSalary AS
3 SELECT emp_name, salary, salary*12 AS AnnualSalary
4 FROM employees;
5
6 • SELECT * FROM EmpAnnualSalary;
7

```

Result Grid

	emp_name	salary	AnnualSalary
▶	Rahul	60000.00	720000.00
	Priya	67000.00	804000.00
	Anil	55000.00	660000.00
	Sneha	70000.00	840000.00
	Kiran	48000.00	576000.00

EmpAnnualSalary 25

Output

Action Output

#	Time	Action	Message
---	------	--------	---------

9. Update an employee's salary through a simple View and verify the base table.

Query 1

```

7 SET salary = 62000
8 WHERE emp_id = 105;
9
10 • SELECT * FROM EmpSalaryView;
11 • SELECT * FROM employees WHERE emp_id = 105;
12

```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	105	Kiran	2	62000.00
*	NULL	NULL	NULL	NULL

EmpSalaryView 26

employees 27

Apply

10. Use SHOW CREATE VIEW to display the definition of a View.

Query 1

```

1 • SHOW CREATE VIEW EmpDept;
2

```

Result Grid

View	Create View	character_set_client
empdept	CREATE ALGORITHM=UNDEFINED DEFINER=`r...` utf8mb4	

Result 28

Read Only

11. List all Views in the database.

```
Query 1 x
1 • SHOW FULL TABLES WHERE Table_type = 'VIEW';
2
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Read Only

Tables_in_employeeodb1	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
deptempcount	VIEW
depthhighavg	VIEW
deptmaxsalary	VIEW
eceemployees	VIEW
empannualsalary	VIEW
empbasic	VIEW
empdept	VIEW

12. Drop a View and verify that it no longer appears in the list.

```
Query 1 x
1 • DROP VIEW IF EXISTS EmpAnnualSalary;
2
3
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Read Only

Tables_in_employeeodb1	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
deptempcount	VIEW
depthhighavg	VIEW
deptmaxsalary	VIEW
eceemployees	VIEW
empannualsalary	VIEW
empbasic	VIEW
empdept	VIEW

Output: Action Output

7. Viva / Discussion Questions

1. What is a View?
2. Why do we use Views?
3. Does a View normally store a separate copy of the data?
4. What is the difference between a View and a table?
5. Can a View be created using multiple tables?
6. Can every View be updated?
7. Why are Views useful for security?
8. What is the difference between CREATE VIEW and CREATE OR REPLACE VIEW?
9. How do you display the definition of a View in MySQL?
10. How do you remove a View?

8. Common Errors and Solutions

Table does not exist: Run USE employeeodb1; and check with SHOW TABLES;

View already exists: Use DROP VIEW IF EXISTS view_name; or CREATE OR REPLACE VIEW.

Unknown column: Check the column names using DESCRIBE employees; or DESCRIBE

departments;

Cannot update View: The View may contain JOIN, GROUP BY, aggregate functions or another non-updatable construct.

Wrong database: Check the selected schema and run SELECT DATABASE();.

9. Quick Reference

```
CREATE VIEW view_name AS
SELECT ...;
SELECT * FROM view_name;
CREATE OR REPLACE VIEW view_name AS
SELECT ...;
SHOW CREATE VIEW view_name;
SHOW FULL TABLES
WHERE Table_type='VIEW';
DROP VIEW IF EXISTS view_name;
```